From Grammar Resources to Interpretable Knowledge Components: An Auditable Dataset and Selection Pipeline

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Abstract

Knowledge tracing (KT) requires an item–knowledge-component (KC) mapping, yet grammar resources usually express targets in prose and do not determine the right KC granularity. We present Grammar–KT, an executable pipeline that converts resource descriptors into canonical grammar cells, generates and independently screens a fixed item bank, derives interpretable KC hypotheses from development grammar, and selects among them using development learner evidence. From 139 English Grammar Profile descriptors, 44 complete mappings yield 24 unique six-dimensional cells and 44 model-validated items covering every cell. The development bank induces 55 feature, operation, interaction, and exact-cell candidates, reduced to 38 activation classes and 28 selection-eligible representatives. A parsimonious forward selector retains nine feature KCs and one perfect–negative interaction. In a 1,600-learner mixed synthetic world, it gives a small fixed-logistic improvement over factorization (Δ log loss $= -.000375$, learner-bootstrap 95% CI $[-.000631, -.000109]$), while its compositional-only interval includes zero. The selected inventory is identical across five full-support simulation seeds; controlled four-world experiments show recovery of strong planted interactions and fallback to marginals under nulls. No representation wins across all latent worlds. We release the dataset, scientific declarations, selection traces, and evidence needed to audit these deliberately limited claims.

1 Introduction

Knowledge tracing (KT) estimates changing learner knowledge from chronological responses (Corbett and Anderson, 1995; Abdelrahman et al., 2023). Its inputs normally associate each item with one or more knowledge components (KCs), often through a binary Q-matrix. That matrix embeds consequential decisions: what counts as reusable knowledge, which interactions deserve separate state, and whether evidence from one grammatical construction should transfer to another. These are scientific choices, not data formatting.

Grammar resources pose a sharp version of the problem. The English Grammar Profile (EGP) contains over 1,200 corpus-derived competence descriptions mapped to CEFR levels (O'Keeffe and Mark, 2017). A description can combine tense, aspect, voice, polarity, clause type, and modality, state bounded alternatives, or fall outside a selected ontology. Treating every descriptor as one KC obscures reuse; decomposing it without explicit rules can invent unsupported distinctions. Even after canonicalization, granularity remains empirical: feature KCs transfer broadly, conjunctions represent dependencies, and exact cells can memorize combinations.

We introduce Grammar–KT, a dataset and research pipeline that makes this path explicit (Figure 1). It first fixes grammar cells and items without access to KCs or learner outcomes. It then constructs a development-only KC hypothesis space from a declared schema and selects supported hypotheses using observable learner history. The resulting policy is frozen before compositional and novel-value probes. This ordering makes leakage, support, and representation complexity inspectable.

We ask four paper-level questions:

RQ1 What dataset results from conservative source normalization and independently screened item generation?

RQ2 What interpretable KC hypothesis space follows from development grammar, and which granularity does predictive–parsimonious selection retain?

RQ3 Does the frozen selection improve prediction on development, compositional, and novel-value probes?

RQ4 How sensitive are selection and prediction to learner support, complexity penalty, KT model, and plausible latent learner worlds?

Our contributions are: (i) a retained medium-scale grammar-KT dataset with a provenance path from descriptors to 44 fixed items and 204,000 chronological events; (ii) a direct candidate procedure spanning reusable features, language-declared operations, supported interactions, and exact cells; (iii) an automated development-evidence selector with a simple predictive–parsimony objective; and (iv) controlled fold, latent-world, support, and validation audits that delimit the contribution. The learners are synthetic and item quality is model-judged, so our results validate method behavior under declared worlds, not cognitive atomicity or classroom efficacy.

2 Related Work

Proficiency-scale grammar resources. The CEFR provides proficiency scales across language activities (Council of Europe, 2020). The EGP specializes this programme for English grammar, using learner-corpus frequency, accuracy, and dispersion to associate competence statements with CEFR levels (O'Keeffe and Mark, 2017). Recent work uses NLP to quantitatively re-examine EGP level assignments (Verratti-Souto et al., 2025). We do not reassess those levels. Our problem is complementary: converting descriptor content into a bounded structural representation while preserving uncertainty and provenance.

Exercise generation and documentation. Adaptive question generation has conditioned generation on learner history and target difficulty (Srivastava and Goodman, 2021; Cui and Sachan, 2023). Grammar-focused systems generate multiple-choice or gap-filling exercises from passages or examples (Chan et al., 2022; Bitew et al., 2023). These studies highlight both the cost of authoring and the need to evaluate correctness and answer uniqueness. We generate common lexical contexts from a fixed grammar cell, screen candidates independently, and freeze the resulting bank before KC construction. Following data-statement principles, we make selection, scope, and limitations explicit (Bender and Friedman, 2018).

Domain models and knowledge tracing. KCs describe acquired units inferred from performance on related tasks (Koedinger et al., 2012). Classical BKT tracks a latent learned/unlearned state with learning, guess, and slip parameters (Corbett and Anderson, 1995); factor models instead use logistic predictors and practice features (Pavlik et al., 2009). A Q-matrix supplies the assumed item–KC mapping, but expert matrices can be refined or learned from response data (Pardos and Dadu, 2018). Learning Factors Analysis explicitly compares and improves cognitive models using learner evidence (Cen et al., 2006), and Picones et al. (2022) combine domain-model induction with student modelling. Our hybrid boundary is different: linguistic structure defines an interpretable candidate space without outcomes; only development responses select within it; and the policy is then frozen. Synthetic KT data support controlled model tests (Piech et al., 2015; Pagonis et al., 2024). We use multiple declared worlds to test recovery and failure modes, not to assert the cognitive truth of any one world.

3 Task Formulation and Source Data

3.1 Grammar-cell representation

For descriptor d, normalization returns a status, cells, explicit Phase-2 eligibility, and a note. A cell has six dimensions,

$$c = (t, a, v, p, q, m), \tag{1}$$

for tense, aspect, voice, polarity, clause type, and modal choice. Table 1 gives the domains. Scalars record exact evidence, lists bounded unresolved alternatives, and null no usable constraint. Dimensions within a cell are conjunctive; separate cells are source-supported alternatives. Lists are not expanded into unsupported Cartesian products.

Mappings are *complete* only when all values are scalar. Partial mappings contain a list or null. Zero-cell statuses distinguish out-of-scope, schema-failure, and unresolved cases. Compatibility constraints include $m \neq \text{none} \Rightarrow t = \text{NA}$ and imperative clauses requiring $t = \text{NA}$ and $m = \text{none}$. Only complete cells enter the downstream canonical inventory.

3.2 Declared source subset

The input is a consult-only parsed EGP snapshot that is not redistributed. Its declared SHA-256 digest and size are checked before execution. From 1,222 records, an ordered list selects 139 descriptors: 23 regression controls and 116 fresh cases

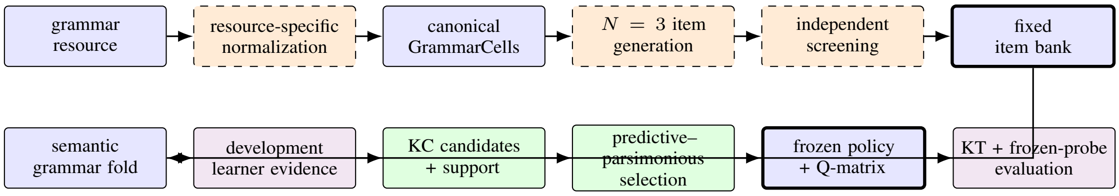

Figure 1: Active Grammar–KT methodology. Model assistance is confined to normalization, item generation, and item screening. Candidate construction uses development grammar only; selection uses development responses only. Thick boxes mark the bank and policy freezing boundaries.

Dimension	Symbol	Closed scalar domain	Principal constraint or interpretation
Tense	t	present, past, NA	NA is structural and differs from missing evidence
Aspect	a	none, progressive, perfect, perfect–progressive	Encodes auxiliary-chain aspect
Voice	v	active, passive	Passive is canonical BE-passive only
Polarity	p	positive, negative	Verbal polarity, not lexical meaning
Clause	q	declarative, polar question, subject-WH, non-subject-WH, imperative	Question subtypes and imperatives are distinct
Modal	m	none; can, could, may, might, must, shall, should, will, would	A central modal requires $t =$ NA

Table 1: The six dimensions of a scalar GRAMMARCELL. During partial normalization, a dimension may instead hold a finite list or null; only scalar cells enter canonicalization.

stratified over tense–aspect combinations, passive voice, negation, questions, imperatives, central modals, and out-of-scope phenomena.

The design contains 139 primary annotation units and eight repeated units. Every unit retains model evidence; only primaries feed canonicalization. The retained annotation snapshot is used as evidence rather than regenerated for this study.

Phase 1 exposes category fields, guideword, and can-do statement. Source examples remain hidden unless a partial result explicitly identifies an uncertain dimension as Phase-2 eligible. Phase 2 may narrow only those domains while preserving every Phase-1 branch and ineligible exact value.

3.3 Fixed dataset boundary and grammar fold

Forty-four complete mappings contribute 48 source–cell edges and 24 unique cells. Item generation and validation see cells and language-facing design files, but no fold, KC, learner outcome, or KT result. After the bank is fixed, a deterministic semantic split assigns 18 cells to development, five to a constituent-compositional holdout, and one containing unseen `modal=would` to a novel-value holdout. The five compositional cells use only feature values seen in development; 36 of their 37 value pairs are also seen. We therefore call the split constituent-compositional rather than claim that every pair has been observed.

4 Grammar-to-KT Pipeline

4.1 Normalization and canonicalization

Each source unit is processed in a fresh model context containing the closed schema, rulebook, phase instructions, and JSON schema. Deterministic validation checks domains, status invariants, and compatibility constraints. Phase 2 is allowed only for explicitly eligible uncertain dimensions; it must narrow a Phase-1 branch, retain every branch, and leave ineligible exact fields fixed. Contradictory or insufficient evidence remains partial or unresolved.

Complete mappings are serialized in a fixed dimension order. A canonical inventory stores each exact tuple once, while a separate edge table preserves every source descriptor and source-cell position. Thus many source statements can support one cell without turning source identifiers into grammatical features.

4.2 Item generation and independent screening

The item generator receives one GRAMMARCELL, the generation prompt, grammar rulebook, and taskformat schema. It chooses common lexical and contextual material and returns an instruction, prompt, answer, and rationale; it receives no KC policy, fold, learner record, or outcome. The default design draws three independent candidates per cell. A deterministic answer-span check first rejects malformed response contracts. A separate model then judges target fidelity, grammaticality, naturalness, pedagogical suitability, determinacy, non-target-language simplicity, answer leakage, extraneous grammar, and world-knowledge dependence. Acceptance requires every criterion.

For each cell we retain the earliest accepted candidate and, when available, the accepted candidate with the greatest token-set distance from it, up to two items. Default $N = 3$ left two cells uncovered. A preregistered unchanged-prompt two-draw rescue covered one; a separate explicit-construction-cue intervention covered the remaining determinacy failure. These cohorts retain distinct provenance and are not relabelled as default-N evidence. An exhaustive agent audit then found five answer-packaging/reference defects and one likely rejected judge error. We corrected only declared non-learner-facing fields, preserved the raw records, and independently rejudged all six corrected candidates with the validator. The resulting 44-item bank is frozen before folding, simulation, or KC construction.

4.3 Development-derived KC candidates

Let D be development cells and their fixed items. Candidate construction is outcome-free and reads only the declared canonical schema, D, the item–cell relation, and a compact design file. It enumerates:

1. observed non-background feature–value KCs;
2. language-specific operations whose activation is deterministically declared from a cell;
3. cross-dimension feature pairs that co-occur in development; and
4. one exact-cell candidate for each development cell.

Background values (e.g., active voice and positive polarity) are explicit English-schema choices rather than hidden program constants. Each candidate stores supporting development cells and items. Pair candidates require at least two cells and three items. We represent every candidate by its binary activation vector over development items and mark identical vectors as one measurement-bank equivalence class, preferring feature candidates as representatives. This is observed-bank equivalence, not a claim of universal linguistic identity. Generic enumeration contains no English feature names; English operations live in a separate declaration.

4.4 Predictive–parsimonious selection

All supported feature–value marginals form the protected initial representation. Remaining nonredundant operation and pair candidates are considered by forward selection. For policy K, a standardized observable PFA-style logistic model (Pavlik et al., 2009) is fitted on development selection-train events and scored on the next chronological development pass. Its features contain only strictly prior learner successes and failures for the currently active KCs; simulator difficulty, latent state, response probability, item identity, and holdout grammar are excluded. We minimize

$$J(K) = \mathrm{LogLoss}_{\text{dev-val}}(K) + \lambda|K|, \tag{2}$$

$$\lambda = 0.0005. \tag{3}$$

At each step the candidate with the strict best objective improvement is added; a final backward pass removes any unprotected addition whose deletion improves the objective. Exact cells are retained as a granularity diagnostic but are not active additions because they do not provide reusable abstractions. The selected activation rules are copied into a plain frozen policy before any grammar-holdout prediction.

4.5 Projection, learner evidence, and KT

For fixed item i and frozen policy K, projection gives

$$Q_{ik} = \mathbb{I}[k \text{ activates on } c_i]. \tag{4}$$

Simulation is policy-independent: a declared latent world first generates one fixed response stream. Each learner receives five shuffled development-only acquisition passes—four for selection training and one for selection validation—followed by one non-updating probe of every bank item. Holdout items therefore provide no acquisition evidence. Policies are compared on identical learners, items, response outcomes, and temporal positions.

We report a smoothed empirical estimator, fixed-parameter multi-KC BKT, and the same observable standardized logistic family. Logistic is primary because a separate audit found BKT's full-credit update to every active KC confounds representation width. Primary logistic excludes simulator-derived difficulty and KC count. Evaluation reports log loss and Brier score primarily, with AUC, calibration error, and accuracy as diagnostics.

5 Experimental Design

Final integrated dataset. The final experiment uses the fixed 44-item bank, semantic 18/5/1 cell fold, the declared mixed latent world, 1,000 learners, and seed 20260827. It contains 204,000 events: 160,000 development acquisition events supplied to selection and 44,000 frozen probes. Four representations reuse exactly this stream: (i) nine factorized feature KCs; (ii) those features plus all seven structurally eligible pair interactions; (iii) the automated policy; and (iv) a labelled oracle with one exact KC for each of all 24 cells. The oracle violates development-only discovery and serves only as a granularity diagnostic.

The primary comparison fixes observable logistic KT and contrasts each policy with factorization. We resample whole learners 5,000 times and report the candidate-minus-factorized change in log loss and Brier score with percentile 95% intervals. Negative values favour the candidate. Development, compositional, and novel-value probe subsets are reported separately. The novel subset has only one cell and two items, so it is descriptive.

Method-development audits. Settings were chosen through staged controls rather than the final test stream. The item audit compares model-selected text with a six-entry controlled lexicon and evaluates $N \in \{1, 3, 5\}$ on eight fixed cells. A 24-item stratified sample is rejudged by the same validator family and an alternate model. The fold audit tests tuple-level coverage and ID/order invariance.

KC robustness uses a fixed 24-cell/42-item structural bank, 240 learners, seeds 20260827–20260829, and four readable worlds: factorized, interaction-heavy, cell-specific, and mixed. We compare factorized, all-supported, automated, and exact-all-cell representations under fixed logistic KT. Nested 30/60/120/240-learner samples study support. A predeclared grid $\lambda \in \{0, .00025, .0005, .001, .002\}$ is evaluated only in the factorized null and interaction-heavy positive control. These synthetic worlds test whether the selector behaves appropriately when its assumptions are true or false; they do not identify a human learner ontology.

For final-bank stability, we rerun selection at 1,000 learners over five mixed- world seeds (20260827–20260831), and on nested 60/120/240/500/1,000-learner subsets of the reference stream. Every run supplies development acquisition events only; holdout and reserved test outcomes are absent from selection.

Models and baselines. Retained normalization and generation outputs use `gpt-5.6-sol` at medium reasoning effort; independent item judgments use `gpt-5.6-terra` at medium effort. Outputs, rendered prompts, parsed records, and call settings are retained. The provider did not expose a pinned snapshot or monetary price. KT baselines are empirical rate, fixed BKT, and regularized logistic; no neural KT model is added because the research variable is KC representation rather than predictive architecture.

After freezing the dataset, a separate 905-call matched operational audit compared medium, high, and xhigh effort with two condition-blind research-agent reviews. A zero-critical-error rule was inconclusive for every stage. For future pipeline runs, high was retained for normalization (higher semantic quality and repeatability than medium, cheaper than tied xhigh), while medium was retained for generation and validation because deeper effort brought no independently confirmed quality gain. This post-construction audit changes the active runner declaration, not the all-medium provenance or results reported for the fixed dataset.

6 Results

6.1 Dataset construction (RQ1)

The final normalization outcome is 44/139 complete mappings (31.7%), 77 partial, 16 out of scope, and two unresolved. The complete rows contribute 48 source–cell edges to 24 unique cells. Explicit Phase-2 eligibility identifies only 9 of the 80 historical partial Phase-1 rows; the retained broad routing had resolved only two. All 80 retained transitions pass the stronger branch-preservation audit, and six adversarial tests reject unsafe changes.

Initial default $N = 3$ generation produced 72 attempts, 71 payloads, 53 validator acceptances, and coverage of 22/24 cells; its initial $N = 1, 2, 3$ prefixes covered 18, 21, and 22 cells. The unchanged-prompt rescue accepted one of four candidates and reached 23 cells. Two explicit-construction-cue

Mapping outcome	Count	Share
Complete	44	31.7%
Partial	77	55.4%
Out of scope	16	11.5%
Unresolved	2	1.4%
Source–cell edges	48	–
Unique cells	24	–

Item cohort	Attempts	Payloads	Accepted	Cells
Curated $N = 1$ prefix	24	24	16	16
Curated $N = 2$ prefix	48	48	33	19
Curated $N = 3$ prefix	72	71	51	22
Unchanged-prompt rescue	4	4	1	1
Explicit-cue intervention	2	2	2	1
Final curated	78	77	54	24
Selected fixed bank	–	–	44	24

Table 2: Source normalization (left) and item construction (right). Prefix rows reuse the same first N default draws after the frozen six-item correction/rejudgment. Initial pre-correction coverage was 18/21/22 cells. Rescue/intervention cohorts have separate provenance; accepted denotes model, not human, validation.

attempts initially accepted one and completed coverage.

An exhaustive agent audit then identified five packaging/reference defects in the selected bank and one likely judge error among rejected items. We corrected only answer/reference fields, preserved the raw artifacts, and independently rejudged this frozen six-item cohort. Two former accepts failed and the former reject passed. Post-correction $N = 1, 2, 3$ prefixes therefore contain 16, 33, and 51 accepts covering 16, 19, and 22 cells; both explicit-cue candidates pass. The curated total is 54 accepts and 44 selected items over all 24 cells. Determinacy fails on 22/77 curated judgments and remains the dominant rejection; naturalness and pedagogical suitability fail on two and three. The other six criteria have no failures. These are model judgments and an agent audit, not expert gold labels.

The earlier eight-cell audit supports the chosen generation design: model- selected language covered 8/8 cells at $N = 3$, versus 5/8 for a controlled lexicon; $N = 5$ added accepted variants but no model-selected cell coverage. On a stratified 24-item reliability sample, original versus same-model repeat acceptance agreed on 19/23 valid rejudgments ($\kappa = .652$), and original versus alternate model on 19/24 ($\kappa = .583$). The final bank therefore supports traceable model-validity claims only.

6.2 Candidate space and selected granularity (RQ2)

The 18 development cells and 32 items induce 55 candidates: nine feature values, ten declared operations, 18 supported co-occurring pairs, and 18 exact development cells. Seventeen candidates duplicate another activation vector, leaving 38 classes; support and equivalence rules leave 28 selection-eligible representatives. Direct negation, passive, imperative, and inversion operations alias corresponding feature KCs on this bank. Finite-tense, perfect-dependency, and progressive-dependency operations remain distinct. Using only one item per cell would leave two eligible interactions rather than seven, showing that item support affects the measurable hypothesis space even when cell support is fixed.

The forward selector starts with nine features and adds only `aspect=perfect∧polarity=negative`, supported by two cells and four items. The frozen automated policy therefore has ten KCs and mean row width 2.18, compared with 16 KCs/2.89 for all supported interactions and 24 unit-width oracle cell KCs.

6.3 Fixed-event KT comparison (RQ3)

On the final mixed-world stream, automated selection lowers all-probe logistic log loss from .643731 to .643356 ($\Delta = -.000375$, 95% learner-bootstrap CI $[-.000631, -.000109]$). The gain is concentrated in development probes ($-.000450$ $[-.000758, -.000141]$); its compositional estimate remains inconclusive ($-.000234$ $[-.000836, .000375]$). All seven supported interactions reach .643334 and improve over factorization on all probes ($-.000397$ $[-.000782, -.000026]$) and compositional probes ($-.001168$ $[-.002042, -.000246]$). This richer policy was not selected by the penalized development objective; its one-stream result is a sensitivity, not grounds to replace the active method. The 24-cell oracle performs worse overall (.657507) and especially on compositional probes because its evaluation-cell columns have no acquisition history.

Novel-value estimates use only 2,000 events from two items. Automated minus factorized log loss is $+.000119$ $[-.000099, .000352]$, and a development- derived policy cannot create a `would`

Candidate family	Raw	Support	Duplicate	Eligible
Feature value	9	9	0	9
Operation	10	7	6	3
Interaction	18	8	2	7
Development cell	18	18	9	9
Total / classes	55	42	17	28

Policy	KCs	Inter.	Q density	KCs/item
Factorized	9	0	.232	2.09
Automated	10	1	.218	2.18
All supported	16	7	.180	2.89
Oracle all-cell	24	0	.042	1.00

Table 3: Development-derived candidate space (left) and frozen policy granularity on the 44-item bank (right). Duplicate means activation-equivalent on development items; "oracle" includes evaluation-cell KCs unavailable to the selector.

Representation	KCs	All-probe LL	Δ all probes [95% CI]	Δ compositional [95% CI]
Factorized	9	.643731	reference	reference
Automated	10	.643356	$-.000375$ [$-.000631$, $-.000109$]	$-.000234$ [$-.000836$, .000375]
All supported interactions	16	.643334	$-.000397$ [$-.000782$, $-.000026$]	$-.001168$ [$-.002042$, $-.000246$]
Oracle all-cell	24	.657507	$+.013777$ [.012288, .015234]	$+.059615$ [.053390, .065675]

Table 4: Primary fixed-logistic results on the final 1,000-learner mixed-world stream. Deltas are candidate minus factorized log loss; intervals resample whole learners 5,000 times. Negative is better. This is one simulation seed.

KC after seeing the holdout. The gap is reported rather than repaired using evaluation grammar.

6.4 Robustness and support (RQ4)

At 240 learners and $\lambda = .0005$, the selector adds no KC in any of three factorized or cell-specific null seeds, and recovers both eligible planted interactions in all three interaction-heavy seeds; one extra operation appears once. At 30 learners the factorized null has additions in 2/3 seeds, while the cell-specific null becomes clean only at 240. Addition-only Jaccard remains .778 in the interaction world and .333 in the mixed world, so repeated events do not eliminate limited structural support.

Penalty sensitivity exposes the intended tradeoff. With $\lambda = 0$, all three factorized-null runs add irrelevant KCs; .00025 does so in 2/3. Penalties .001 and .002 miss at least one planted interaction in 1/3 and 3/3 positive-control runs. The retained .0005 is the only tested point with zero null additions and joint 3/3 planted recovery. Across worlds, automated minus factorized mean log loss is 0 in the factorized world, $-.002256$ in the interaction-heavy world, 0 in the cell-specific world, and $+.000016$ in the mixed world. Exact cells win only in their matching cell-specific world. No representation is universally best.

On the curated final bank, all five 1,000-learner mixed-world seeds select the same ten KCs, including perfect$\wedge$negative (pairwise inventory Jaccard $= 1$). Nested 60/240/500/1,000-learner subsets of the reference seed also match; the 120-learner subset instead adds present$\wedge$passive, giving Jaccard .818 with the reference inventory. Thus full-support selection is stable across the tested simulation seeds, while a single intermediate support sample exposes finite-evidence sensitivity.

7 Discussion

RQ1: conservative construction trades coverage for traceability. Only 31.7% of sampled descriptors support exact cells, but partial and out-of-scope cases remain explicit rather than being forced into the ontology. Best-of-three generation covers most, not all, cells; the remaining failures reveal a specific weakness—contexts often fail to determine complex perfect forms. Separately recorded rescue and explicit-cue cohorts make that failure-driven intervention visible. The post-generation audit and independent rejudgment also show that accepted payloads can hide answer-packaging errors. Forty-four unique prompts across all 24 cells are a usable research bank, but model acceptance is not human naturalness or pedagogical efficacy.

RQ2: selection supports conditional, not universal granularity. The candidate space operationalizes the continuum from transferable feature values to exact cells. Support and activation equivalence remove many apparent choices before outcomes are inspected. On the final stream, automation adds one interaction to nine marginals. Phase-5 controls give the stronger reason to retain the method: it recovers large reusable dependencies when planted and falls back under two null structures. Exact cells expose its designed blind spot, while the mixed world shows that weak interactions can remain uncertain. The selected policy is therefore an empirically calibrated hypothesis, not a unique cognitive decomposition.

RQ3: a small within-stream gain does not establish transfer. The final automated policy improves aggregate and development-probe prediction over factorization in the retained stream, but its compositional-only interval includes zero and its novel-value point estimate reverses sign. The unselected all-supported policy improves compositional probes in this stream, whereas the Phase-5 worlds show that the same added complexity is not consistently useful. The selector detects strong interactions, but the final mixed dataset does not establish that its chosen interaction improves compositional transfer.

RQ4: support and latent assumptions dominate robustness. Full-support selection is identical across five final-bank simulation seeds, but the 120-learner nested condition selects a different interaction and the smaller Phase-5 bank remains less stable. Rankings reverse between interaction-heavy and cell-specific worlds. BKT also favours wider representations under its shared full-credit update. We consequently fix one observable logistic selector, report learner-paired effects, and treat world, penalty, evidence volume, and KT family as sensitivities rather than searching for a globally best configuration.

Language-generality boundary. Candidate enumeration, activation, support, equivalence, forward selection, freezing, and projection consume arbitrary declared dimensions and values. An alternate toy schema using mood and person passes this interface end to end. The six dimensions, their background values, normalization prompts, and operation declarations are English-specific. The test establishes structural reuse of software, not empirical cross-lingual validity or a universal grammar ontology.

8 Reproducibility and Audit Trail

Researcher-facing YAML/text files separately declare the English schema, normalization boundary, item prompt and criteria, semantic fold, latent worlds, candidate families and support, selector penalty, and KT protocol. Direct Python procedures materialize each transformation. The retained dataset contains source rows, mappings, canonical edges, all 78 generation attempts, 77 original judgments, six correction judgments, the 54-item accepted pool, 44-item selected bank, fold, 204,000 events, private oracle, 55-candidate inventory, selection trace, five-seed stability study, frozen policies, Q-matrices, predictions, metrics, and paired comparisons.

The executable sequence is `run_full_dataset.py` (default, rescue, then intervention), `curate_item_packaging.py`, `finalize_full_dataset.py`, `run_phase6_selection_stability.py`, and `analyze_full_dataset.py`. Exact invocations—including both model names, medium reasoning effort, four workers, seed 20260827, 1,000 learners, and 5,000 bootstrap replicates—are retained in the dataset manifests and append-only experiment ledger. Large event and prediction files use deterministic gzip metadata. A no-live-call notebook executes the same active functions. The external EGP snapshot is consult-only, which limits independent source-stage replay; derived rows and declared source metadata remain auditable in the package.

Future live runs read per-stage model and effort choices from the compact `modules/model_backends.yaml` declaration. The matched effort-audit manifest retains its 905 prompts/outputs, token and latency records, blind reviews, adjudications, paired resamples, strict inconclusive outcomes, and operational high/medium/medium decision separately from the final dataset.

9 Conclusion

Grammar–KT turns grammar-resource evidence into a fixed, model-screened item bank and an interpretable KC selection problem. Development grammar yields feature, operation, interaction, and exact-cell hypotheses; structural support and activation equivalence reduce them before an observable predictive–parsimony selector freezes a policy. Controlled worlds show that the selector can recover strong supported interactions and reject them under null structures, while the final mixed-world gain is small, compositional transfer remains unresolved, and no granularity is universally best. The resulting dataset, active pipeline, negative results, and claim boundaries provide an executable methodology for future expert and human-learner validation rather than a claim that synthetic evidence settles the cognitive ontology of grammar.

Limitations

The empirical grammar domain is English single-clause verbal morphosyntax. The 139-descriptor sample is not the full EGP, only exact mappings enter downstream stages, and the retained normalization snapshot has eight repeated units but no expert adjudication or broad cross-model stability study. The external parsed snapshot is not redistributed. The alternate-schema test verifies an interface, not another language.

Items are LLM-generated and model-validated. The generator and validator are different model configurations, and a repeat/model sensitivity study reveals material disagreement; no teachers or learners assessed correctness, naturalness, difficulty, accessibility, or educational value. The final explicit-construction cue was introduced only after a preregistered unchanged- prompt rescue failed, so its distinct provenance is essential. Provider model snapshots, complete decoding controls, and prices were unavailable.

The exhaustive item audit was performed by an independent agent, not a teacher or annotation panel. Its six frozen corrections were rejudged by the same validator configuration used for the bank, so the resulting curation removes identified packaging defects but does not provide independent human validity.

All learner responses are synthetic. The four worlds omit vocabulary state, forgetting, transfer among KCs, response strategies, classroom exposure, and many forms of individual variation. The final integrated result uses one mixed-world seed; learner bootstrap quantifies clustering within that stream, not metric uncertainty across latent worlds or simulation seeds. Five full- support mixed-world seeds produce the same selected inventory, but this is stability under one declared simulator family, not replication across human populations, item-generation runs, or model versions.

Candidate equivalence is item-bank-specific. Background values, protected marginals, two-cell/three-item pair support, and exclusion of exact cells from active additions are explicit assumptions. Repeated item variants change which pair candidates meet item support. Observable PFA gives each active KC full response credit, while multi-KC BKT has a stronger shared-credit confound. Novel-value evaluation contains only one cell, and a development-derived policy cannot label its unseen `would` value without violating the holdout. None of the results establishes cognitively atomic KCs or human compositional learning.

Ethics Statement

No learner or human-participant records are used; all response sequences are synthetic. Latent-world labels denote simulation assumptions, not demographic or ability groups. Generated items must not be deployed for assessment or instruction without qualified human review for correctness, ambiguity, accessibility, and bias. Use and quotation of EGP material remain subject to its licence; the package retains derived mappings and verification metadata rather than redistributing the source snapshot.

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A Candidate and Operation Audit

The active English background values are tense `NA`, aspect `none`, active voice, positive polarity, declarative clause, and modal `none`. Present and past remain explicit KCs. Making every observed value explicit expanded the Phase-2 raw space from 48 to 107 candidates and produced 53 duplicate excess candidates, so the retained backgrounds are a parsimonious declaration rather than a cognitive claim.

Of ten English operation declarations, finite-tense form, perfect dependency, and progressive dependency are nonredundant and selection-eligible. Negation, passive dependency, imperative, and operator inversion duplicate corresponding feature columns. Central modal, subject-WH, and WH-fronting have zero development support. Generator self-reported operation tags are not used. Candidate construction has no learner-event or holdout parameter; mutating all holdout features leaves the retained development inventory byte-identical.

With one rank-1 development item per cell, the same 55 raw candidates yield 23 eligible representatives and two eligible interactions. With up to two items, they yield 28 and seven. The five newly eligible pairs all meet the unchanged two-cell requirement but gain the required third item. This is a structural sensitivity; outcomes were not read and selection was not rerun.

B Selector Sensitivity and Latent Worlds

At 240 learners, selected KC counts across the three seeds are 9/9/9 in the factorized world, 11/12/11 in the interaction-heavy world, 9/9/9 in the cell-specific world, and 9/9/10 in the mixed world. On the curated final bank, five independent 1,000-learner mixed-world streams select the identical ten-KC inventory (all ten pairwise Jaccards equal 1). In nested reference-seed samples, 60, 240, 500, and 1,000 learners match that inventory; 120 learners substitute present$\wedge$passive for perfect$\wedge$negative (Jaccard .818). No holdout or reserved test event is supplied to any selection run.

On the reference interaction-heavy seed, automated minus factorized all-probe log loss is $-.002666$ with learner-bootstrap interval $[-.004244, -.001103]$; the compositional-only interval $[-.005286, .002518]$ includes zero. This separates predictive recovery of a planted interaction from the unresolved transfer claim.

λ	Null runs with additions	Null KC counts	Interaction KC counts	Both planted recovered
0	3/3	14, 14, 11	13, 15, 13	3/3
.00025	2/3	9, 10, 10	11, 13, 12	3/3
.0005	0/3	9, 9, 9	11, 12, 11	3/3
.001	0/3	9, 9, 9	11, 11, 10	2/3
.002	0/3	9, 9, 9	9, 10, 10	0/3

Table 5: Complexity-penalty sensitivity at 240 learners. The null is the factorized world; recovery is measured in the interaction-heavy world over seeds 20260827–20260829.

Latent world	Factorized	All supported	Automated	Oracle cell	Auto.–fact.
Factorized	.578311	.578334	.578311	.603004	.000000
Interaction-heavy	.609353	.607225	**.607097**	.621903	-.002256
Cell-specific	.686715	.686379	.686715	**.679439**	.000000
Mixed	.641559	**.641273**	.641575	.652886	+.000016

Table 6: Three-seed mean frozen-probe logistic log loss on the structural robustness bank. Bold marks the lowest value within a declared synthetic world, not a human-validity ranking.

C Item and Validation Audit

The eight-cell item audit made 89 generation attempts and 86 structurally valid objects. Model-selected $N = 1/3/5$ covered 6/8, 8/8, and 8/8 cells; controlled lexicon generation covered 5/8, 5/8, and 7/8. Model-selected $N = 3$ accepted 18/24 candidates, while $N = 5$ accepted 27/40 without further coverage. On three matched cells, readable source evidence and cell-only prompting each accepted six candidates at $N = 3$; that sample is too small for a source-evidence claim.

The final bank's 44 prompts are all unique. They contain 242 lexical types in 795 lexical tokens (type–token ratio .304). Twenty rank-2 items have median token-set distance .739 from their cell's first item. These automatic surface statistics show nonidentity, not pedagogical diversity.

Before curation, the exhaustive agent audit classified 31/45 selected items as having no material item-level concern, nine as usable but judgment-sensitive, and five as requiring deterministic packaging/reference correction. Six corrected candidates were independently rejudged: two original accepts became rejects, one original reject became accepted, and three accepts remained accepted. The raw records and superseded downstream artifacts are retained. This audit is qualitative agent evidence, not expert or learner validation.

D RQ and Artifact Map

RQ	Status	Evidence-supported answer
RQ1	Partial	24 cells and 44 model-validated items are retained; source completion is 31.7%, and no human item study was performed.
RQ2	Partial	The bank gives 55 raw/38 activation-class/28 eligible candidates; automation freezes nine marginals plus one interaction.
RQ3	Partial	Automation improves the retained stream overall, but its compositional and novel-value intervals include zero.
RQ4	Answered within controls	Recovery, penalty, support, KT, and four-world audits show assumption sensitivity and no universal winner; the final inventory is identical over five full-support seeds.

Table 7: Paper-level research-question ledger. "Partial" marks missing human or cross-world evidence, not an unfinished software stage.

Component	Retained declaration, executor, and evidence
Source & normalisation	Source manifest, prompts, rulebook, schema; `io.py`, `normalise.py`; retained mappings and transition audit.
Canonical & items	English dimensions, generation prompt/design, validation criteria; `canonicalise.py`, `generate.py`, `validate_items.py`; all calls, judgments, accepted pool, and selected bank.
Fold & simulation	Semantic fold, frozen-probe protocol, four world files; `fold.py`, `simulate.py`; fixed events and private oracle.
KC method	`candidate_design.yaml`, English operation declaration, `selection.yaml`; `kc_candidates.py`, `kc_selection.py`; inventory, trace, policies, and Q-matrices.
KT & evaluation	Observable KT protocol; `kt.py`, `evaluate.py`; predictions, regime metrics, and learner-paired comparisons.

Table 8: Compact artifact map. Exact commands, model settings, seeds, hashes, negative results, and output paths are recorded in the experiment ledger.